Lecture 10: Linear Bandits in Practice

ID 6002W: Online and Reinforcement Learning

Web-enabled M.Tech. program, IIT Madras

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The Linear Bandits Model

An extension to multi-armed bandits

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  - each with fixed but unknown rewards μ_i
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  - each with fixed but unknown rewards μ_i
    - No relation between $\mu_1, \ldots, \mu_K$ is assumed
- The linear bandit model adds some structure to $\mu_1, \ldots, \mu_K$
  - $\mu_i = \langle \theta, x_i \rangle$: $x_i \in \mathbb{R}^d$ known, $\theta \in \mathbb{R}^d$ unknown
    - Allows us to use data from one arm to learn about other arms

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Regret depends on K	Dependence shifts to d

Today's Lecture

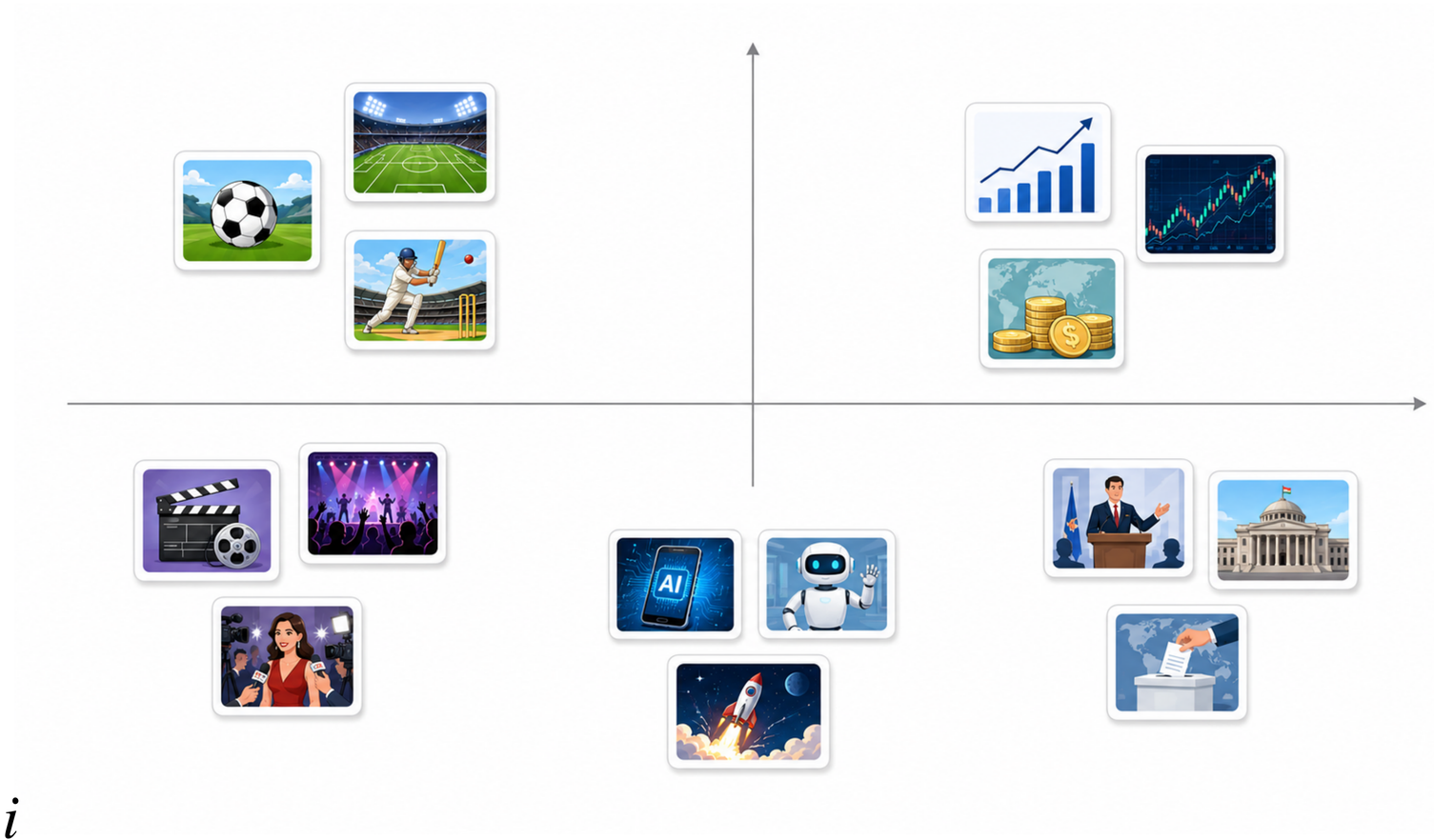
Outline

- What is the interpretation of the context vector x and unknown vector θ?
- How do we obtain the context vector x in practice?
- Can the context set $\mathcal{X} = \{x_1, \ldots, x_K\}$ change with time? How does that change the algorithm?
- Can the model be generalised to make personalised decisions?

This lecture is about the practical aspects of linear bandits

Interpreting Context Vectors

- We know that $\mu_i = \langle \theta, x_i \rangle$
- This means, arms i, j with $x_i \approx x_j$ will have $\mu_i \approx \mu_j$ (for any θ)
- Intuitively, arms that are ‘similar’ should have similar rewards; closer the arms, closer the rewards
- So context vector x_i should capture some semantic meaning about arm i



Representative image of context vectors of different news articles.
Articles of similar topics should have context vectors that are close together

Why Linear Functions?

- The model says $\mu_i = \langle \theta, x_i \rangle$
- In other words, $y(t) = \langle \theta, x(t) \rangle + \eta_t$
- This model gives us a closed-form solution for $\hat{\theta}$
  - The solution can be easily iterated at each time t
- More importantly, it gives us a confidence set with theoretical guarantees:
  - With probability at least $1 - \delta$, $\|\theta - \hat{\theta}\|_{V_t} \leq \beta$

Is the reward truly a linear function of the arm features?

Is Linearity Too Restrictive?

- The model says $\mu_i = \langle \theta, x_i \rangle$
  - **But x_i need not be the raw item features**
- Raw input (article text, image, user profile) is passed through a feature map φ

$$x_i = \varphi(\text{raw item } i)$$

- φ must be **fixed before** we run the bandit algorithm
  - The bandit learns θ, not φ
- Richer $\varphi \Rightarrow$ larger d. More expressive, but slower learning (larger regret)

The real question is, how to choose φ?

Where Do Context Vectors Come From?

Where Do Context Vectors Come From?

Let us understand with the example of news article recommendation

Hand-Crafted Features

Coarse Features

- Each news article has a topic/category:
  - Encode category as one-hot or multi-hot vector
  - d categories $\Rightarrow$ d-dimensional context vectors

✓ Low-dimensional, interpretable features

- Misses subtler differences within a category

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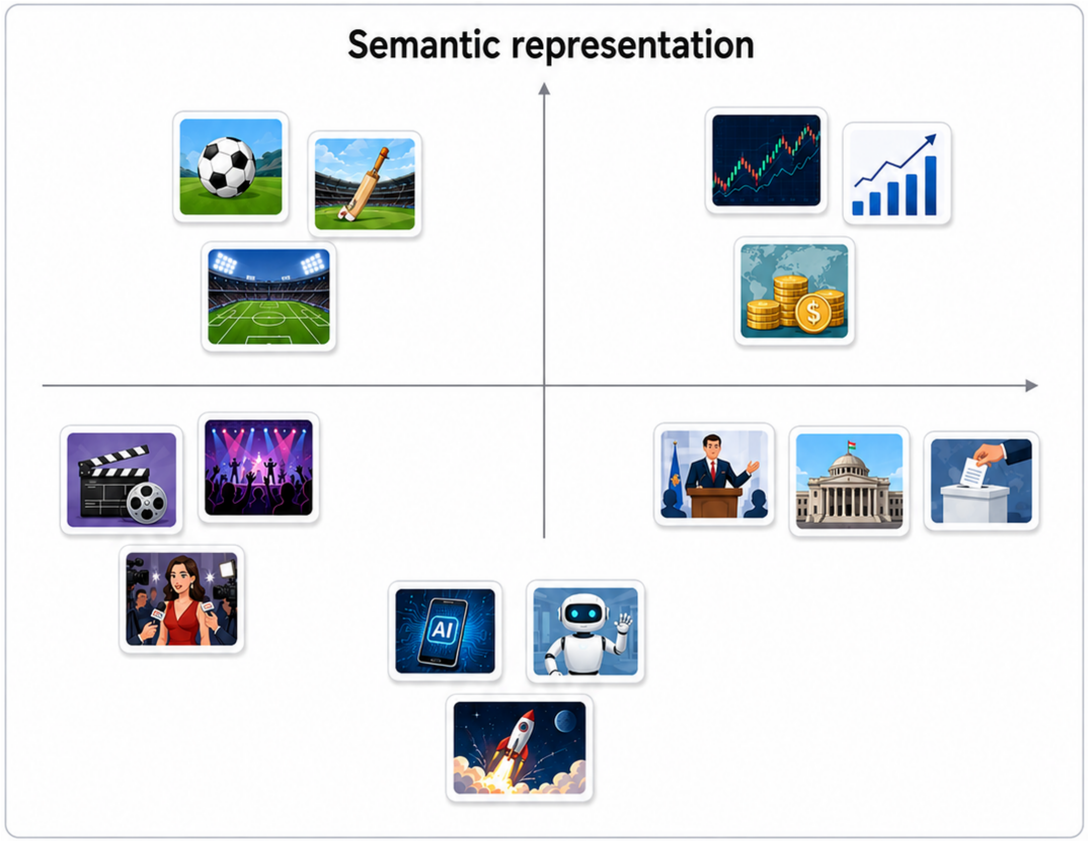
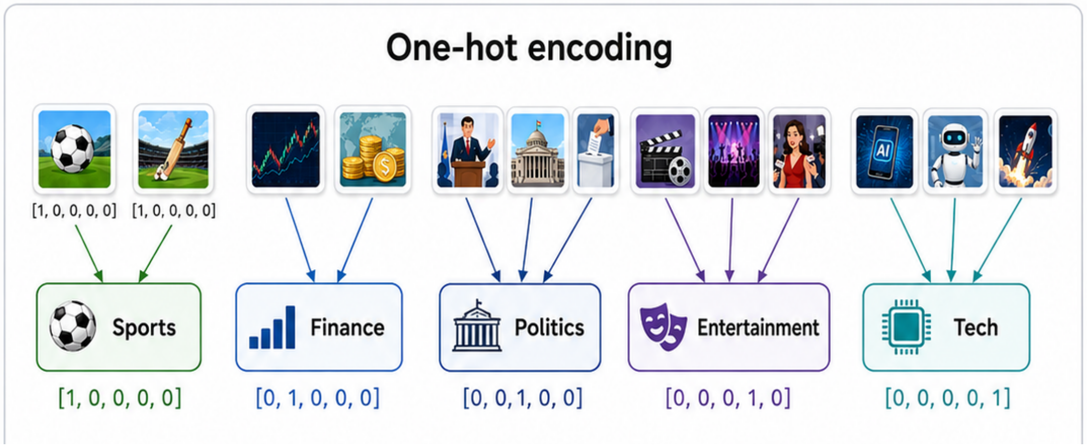




No notion that politics is closer to finance than entertainment

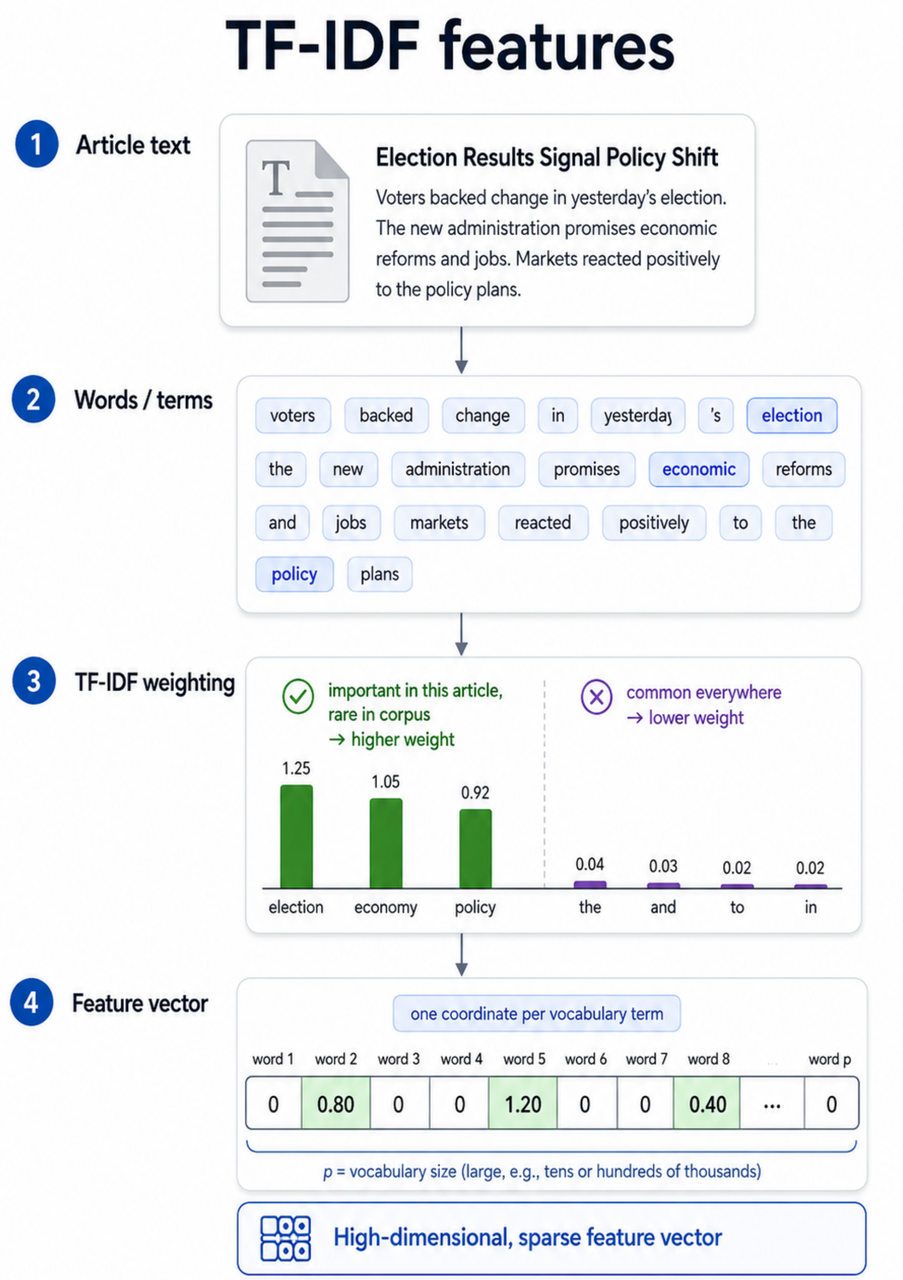
Hand-Crafted Features

Fine Features

- Each news article is a piece of text
  - Use calculate text-based features: bag-of-words, TF-IDF, BM25, etc.
  - Each word/term becomes a possible feature

✓ Captures richer item information

- Very high-dimensional ⇒ poor learning



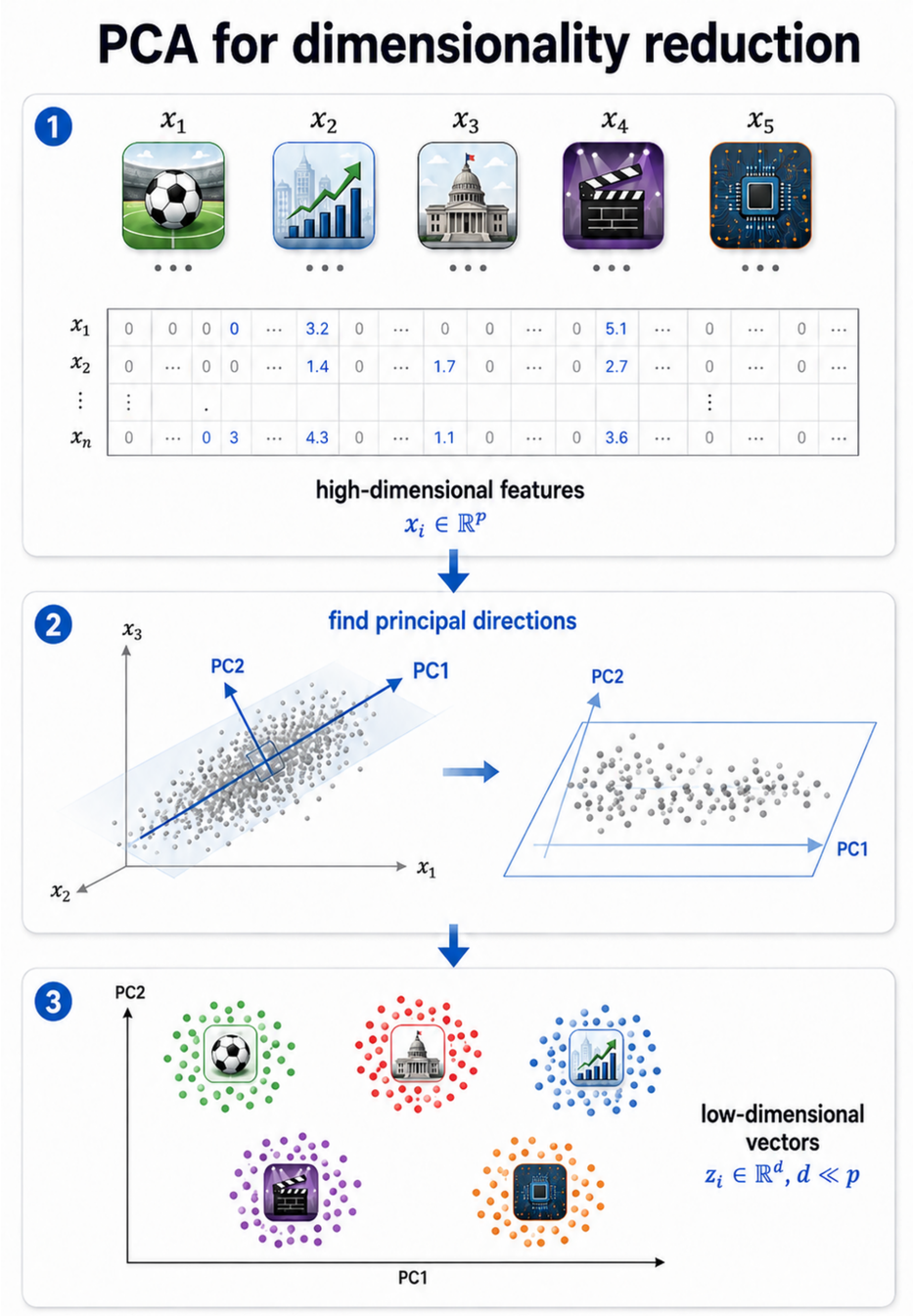
Hand-Crafted Features

Dimensionality Reduction

- Start with high-dimensional item vectors x_i
- We can do dimensionality reduction,
  e.g. via Principle Component Analysis (PCA)
- It takes projections of each x_i onto low-dimensional space, preserving distinctive qualities

✓ Keeps richer signals in fewer dimensions

- Loses direct interpretability of x_i



Neural Network Features

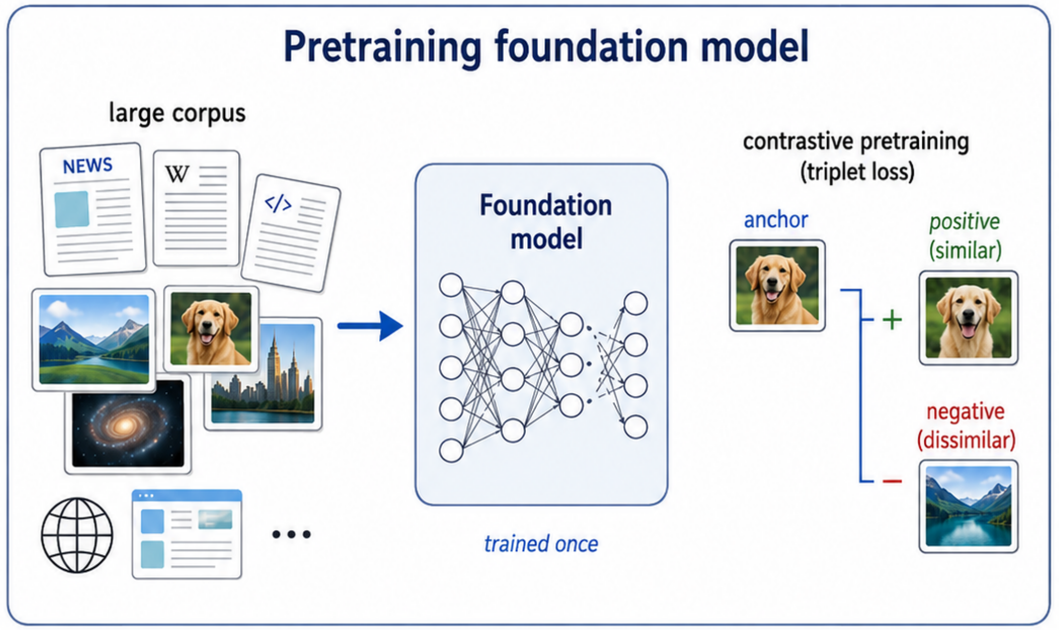
Off-the-shelf foundation models

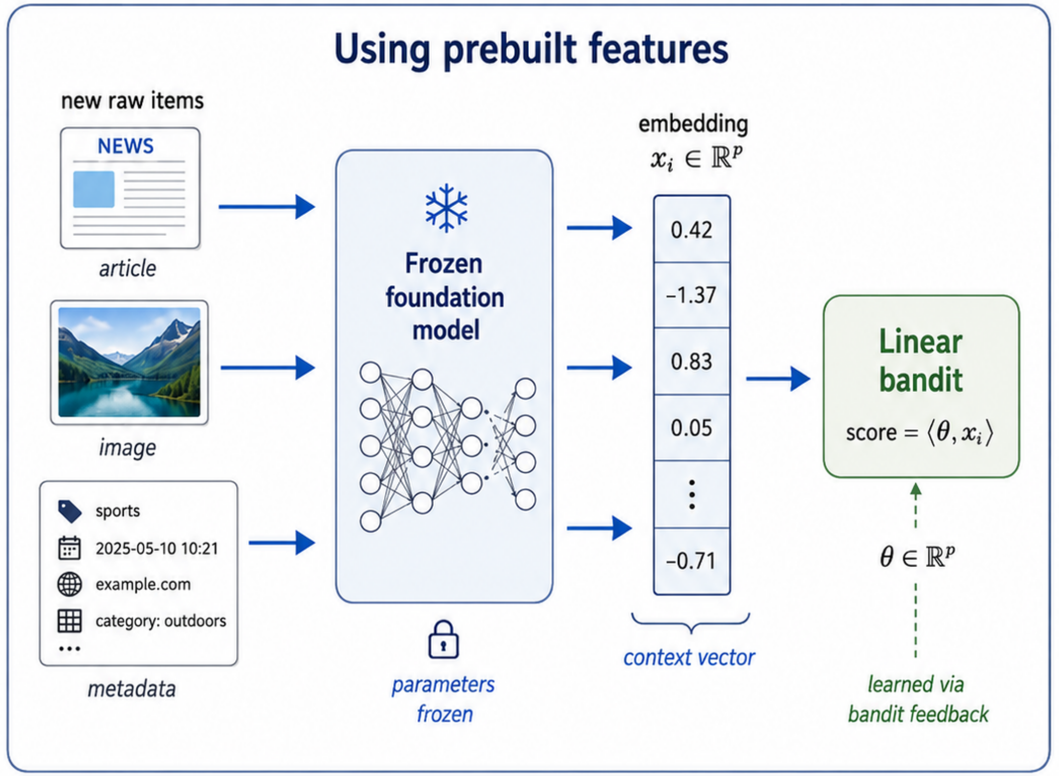
- Foundation models are gigantic neural networks trained on a huge corpus of data

- They learn embeddings that reflect semantic similarity
  - CLIP-style embeddings for images
  - BERT-style embeddings for text

✓ Captures rich semantics from complex data types

- High dimensional features ($d = 768, 1024$), potentially *not aligned* with task at hand





Neural Network Features

Training a neural network ourselves

- We have logged data: which articles were shown, clicked/not clicked

- Train a neural network to predict clicks

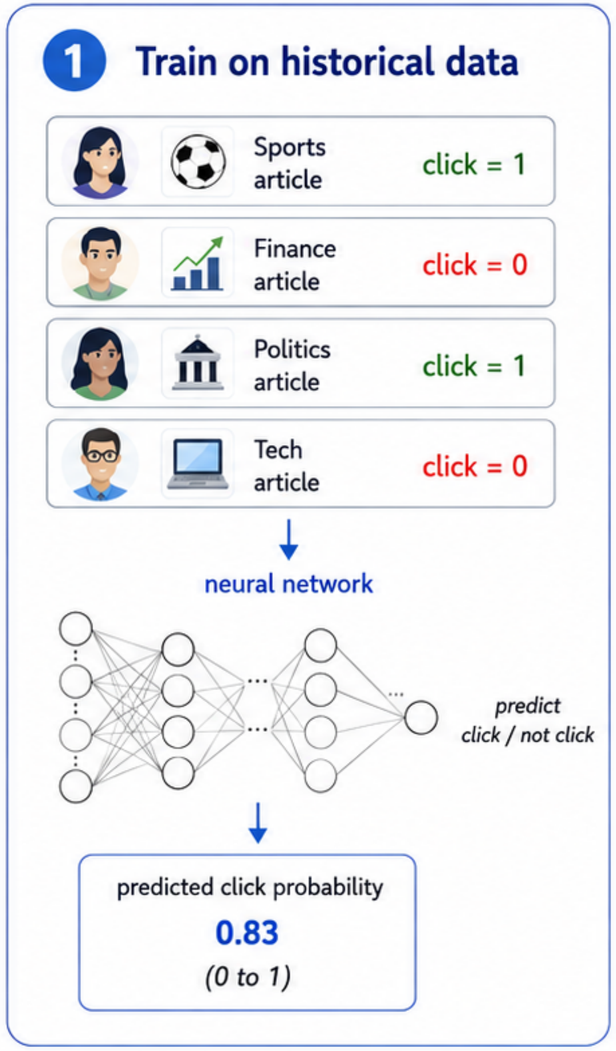
- Remove the last layer and use the rest as φ

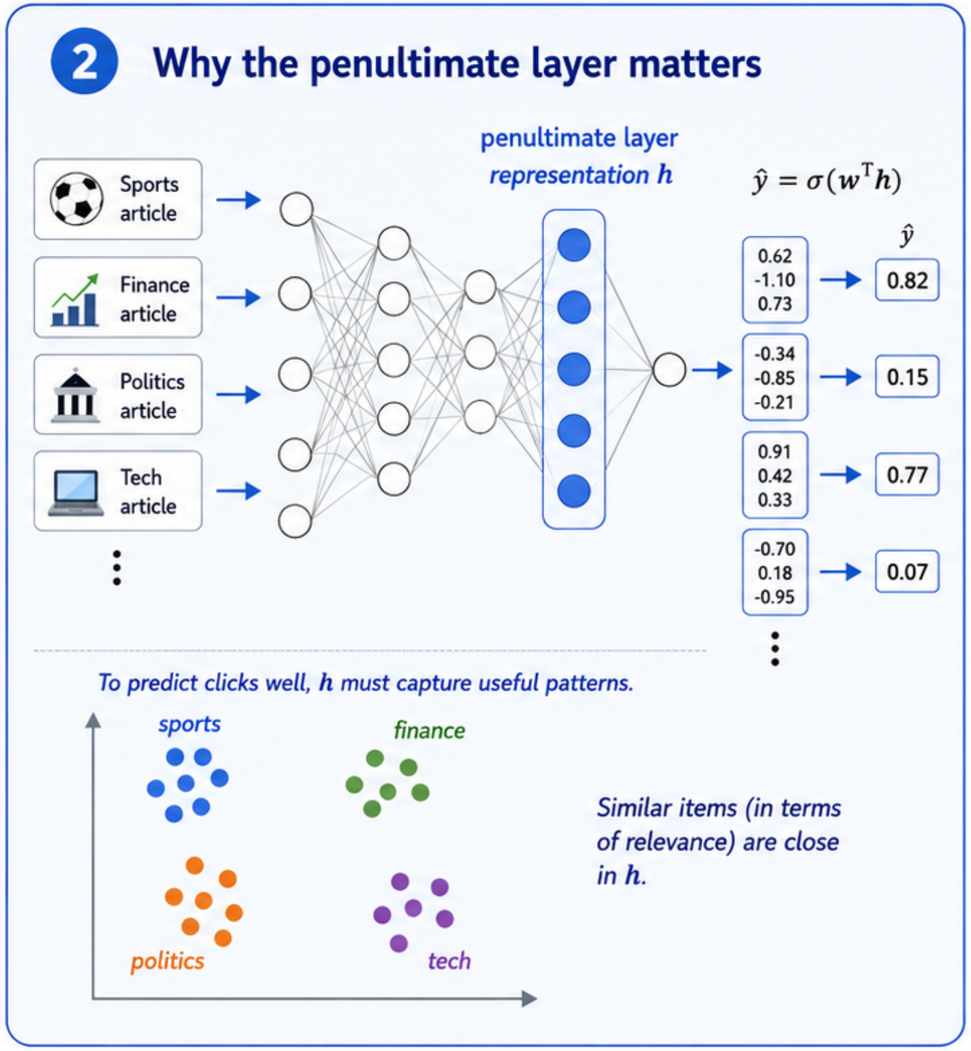
✓ Extracts rich features relevant to the task at hand

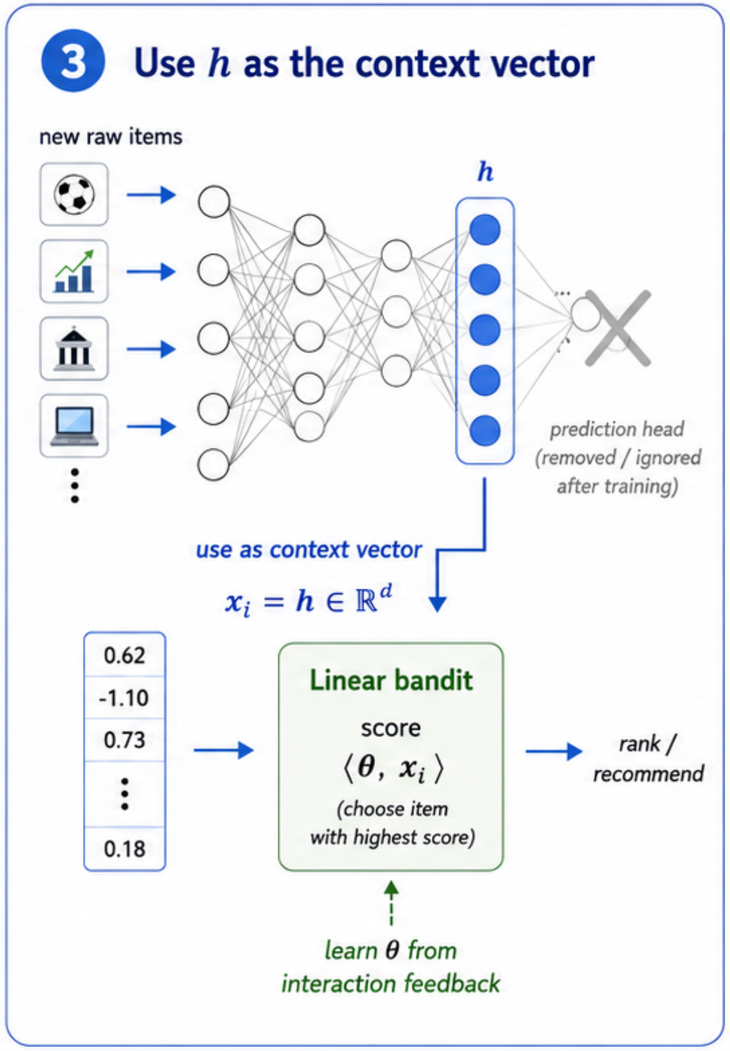
- Needs lots of past data to be effective

Neural Network Features

Training a neural network ourselves







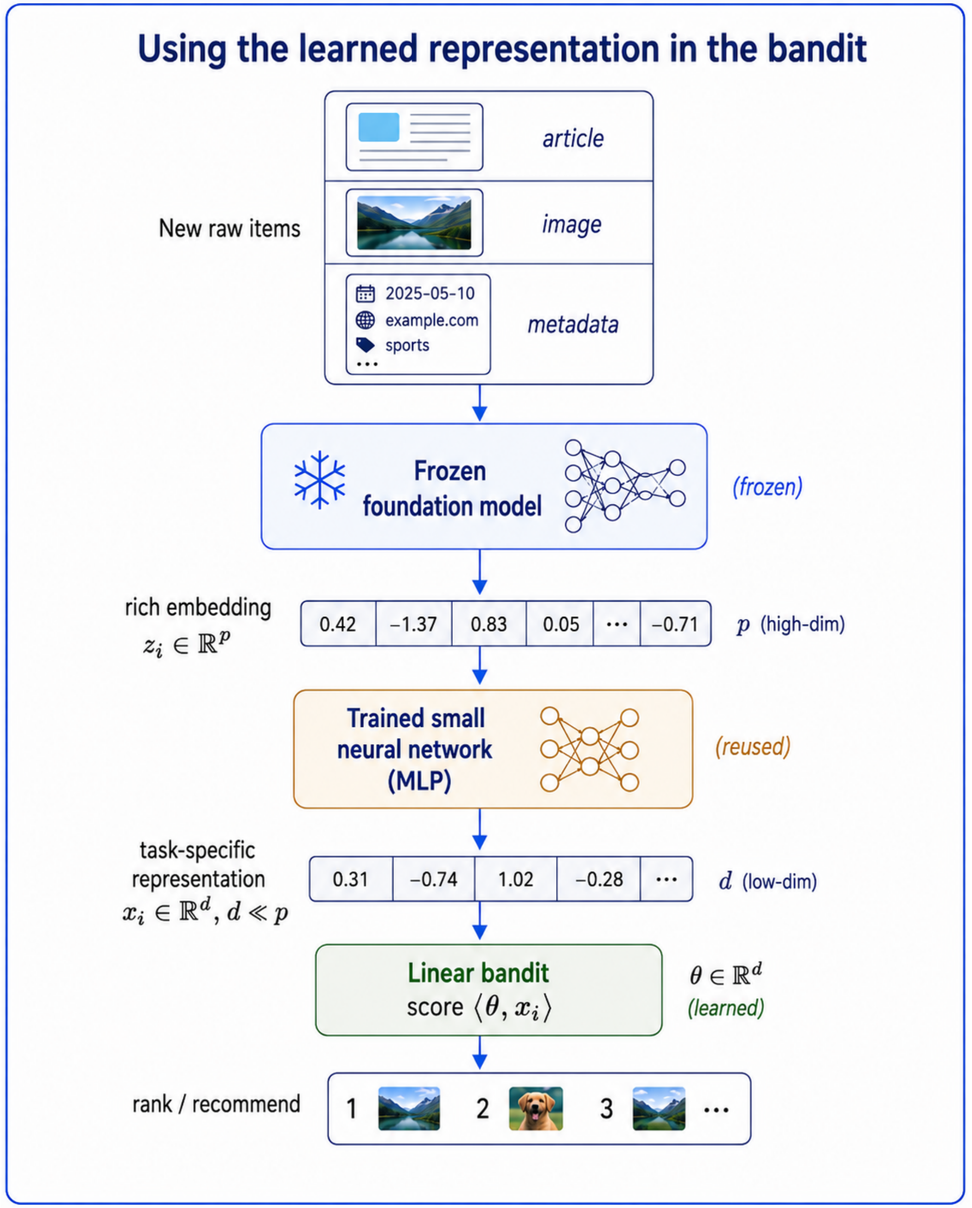
Neural Network Features

Modern Paradigm

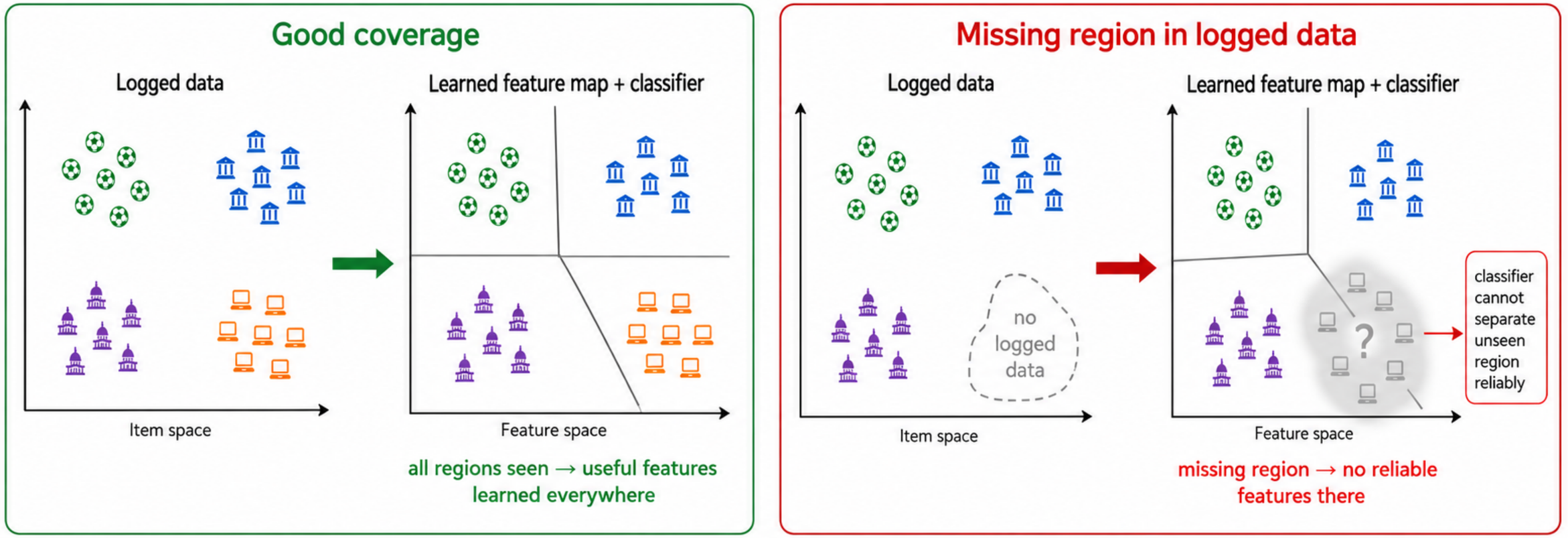
- Use foundation models and custom network, back-to-back
- Foundation model gives rich embedding
- Small custom network maps it to task-specific feature
- Bandit algorithm works with this low-dimensional representation

✓ Rich features, less data hungry

- Still depends on good exploratory logged data



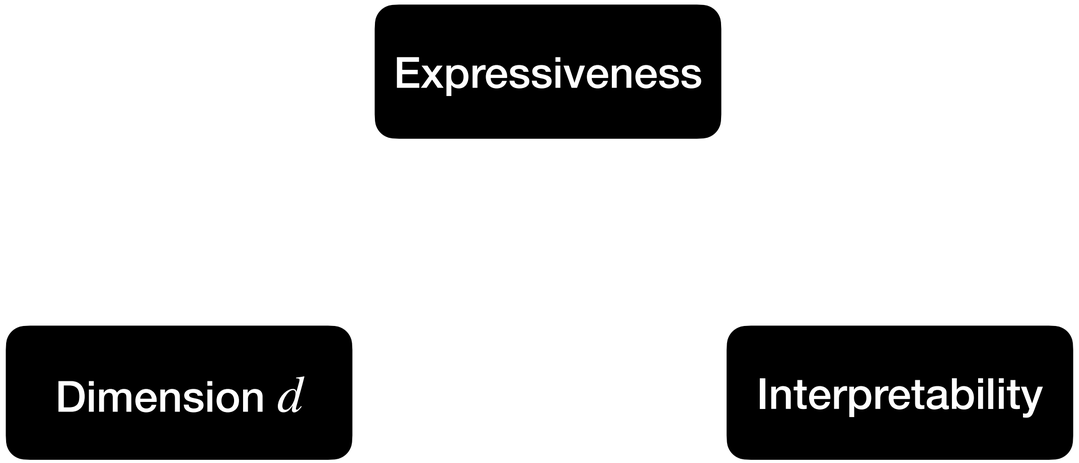
Why Exploratory Logging Matters



If a whole region is never explored in the logged data, the supervised learner cannot learn a good representation for that region.

Final Words on Feature Engineering

Bandit-Specific Caveats



- Feature scaling matters: if some items have large features and others have very small, then the large feature items will automatically win!
-

Extending The Linear Bandit Model

Time-Varying Action Sets

- The set of arms available may not remain fixed throughout
  - News articles: new ones published, old ones retired
  - Products: new ones listed, old ones out of stock
  - Ads on Google: ad pool changes as a function of search

But there are interesting consequences on the learning dynamics

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- The set of arms available may not remain fixed throughout
  - News articles: new ones published, old ones retired
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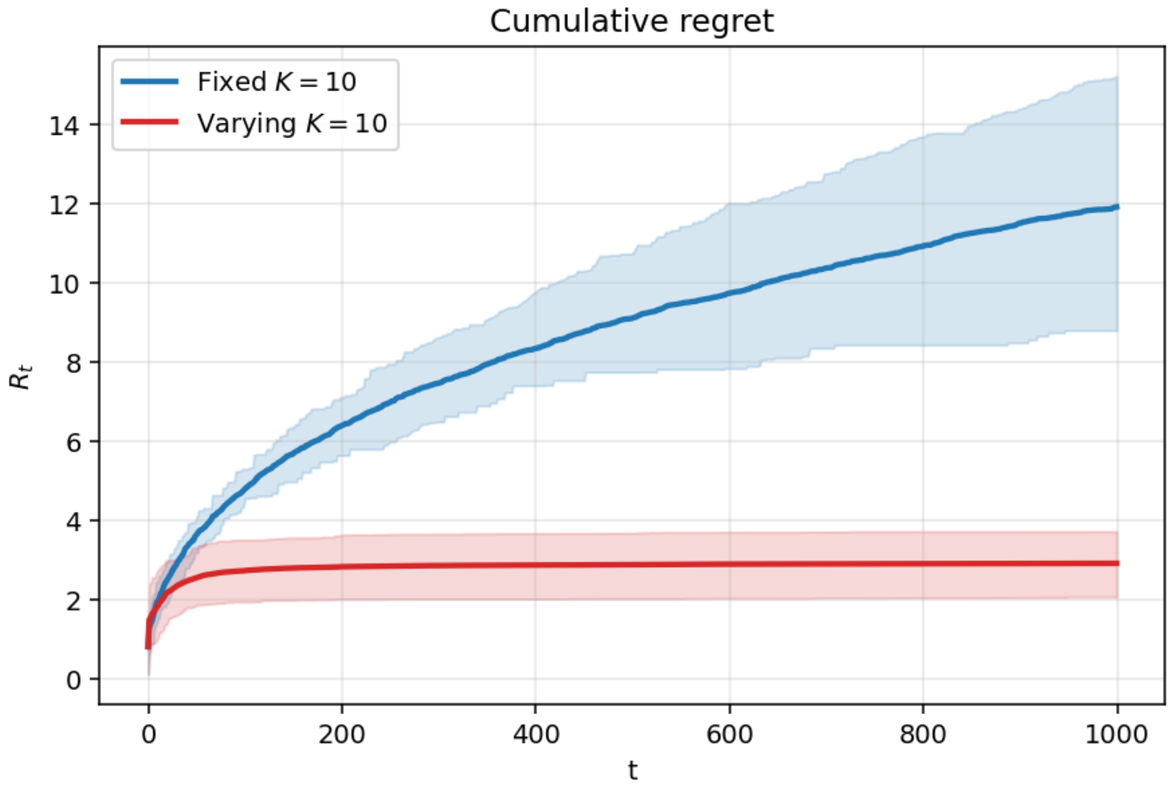
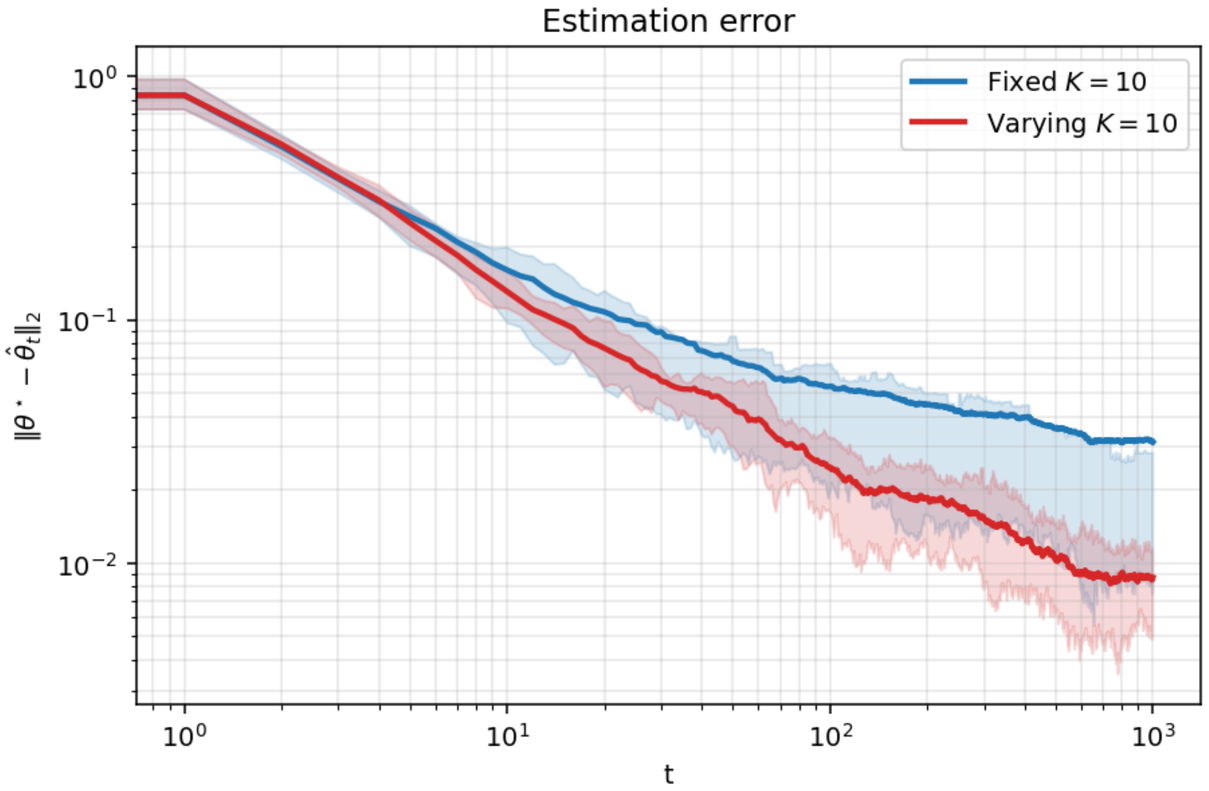
  - Ads on Google: ad pool changes as a function of search
- The linear bandit algorithms we have studied can handle this case!
- Simply compute $\arg\max_{x \in \mathcal{X}_t} \langle \hat{\theta}_t, x \rangle + \sqrt{\beta} \|x\|_{V_t^{-1}} : \mathcal{X}_t = \{x_{1,t}, x_{2,t}, \ldots, x_{K,t}\}$

But there are interesting consequences on the learning dynamics

Regret and Estimation Error

Fixed context versus varying context: K = 10





Confidence Ellipsoids and Arm Scatter Plots

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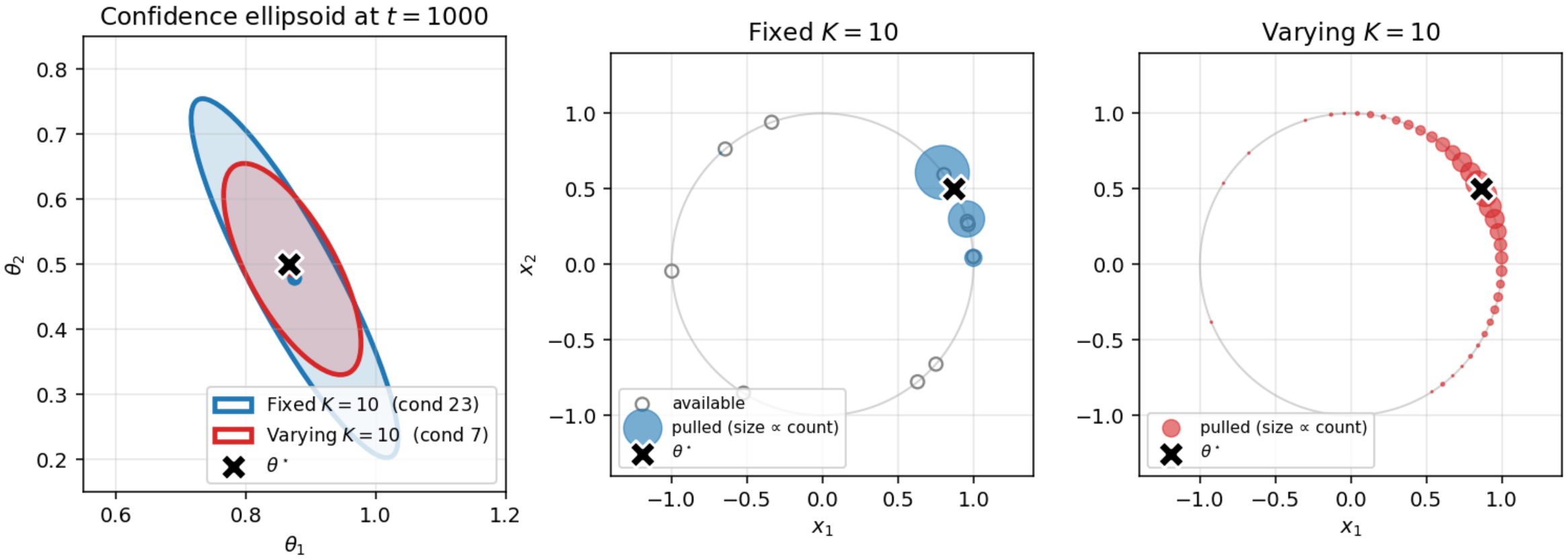

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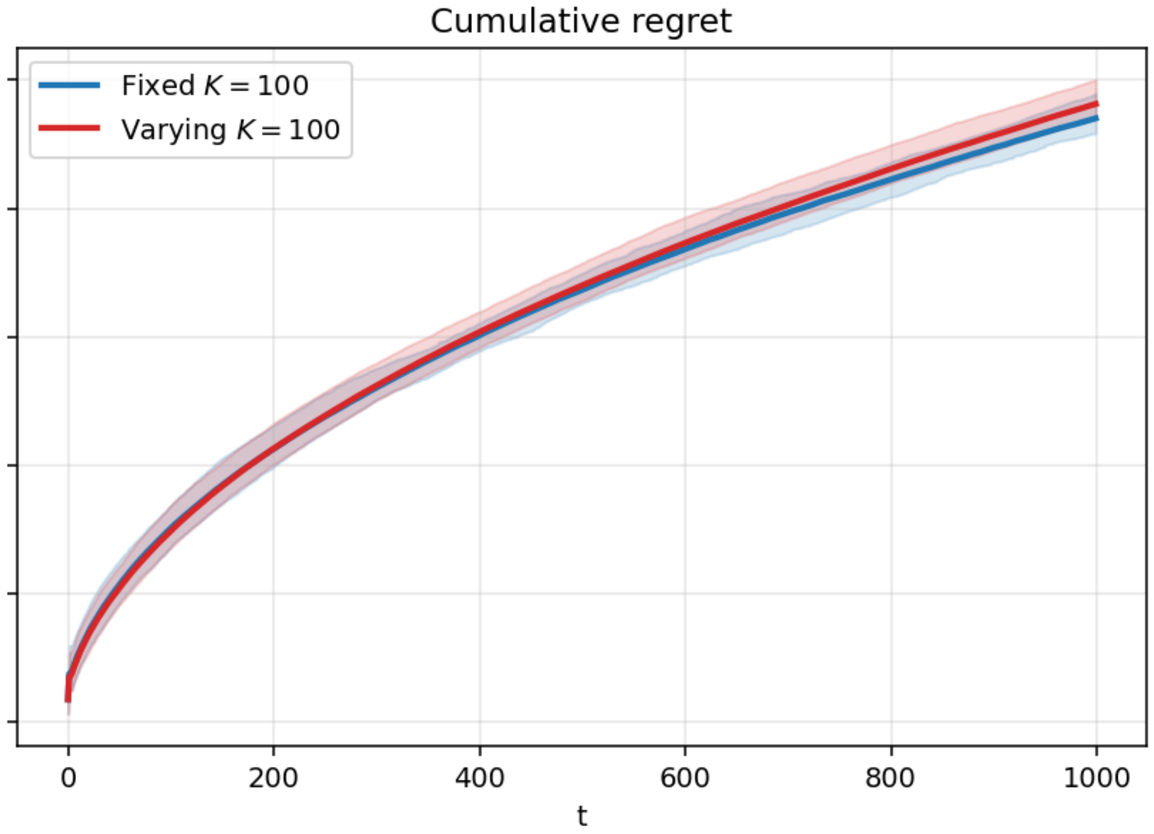

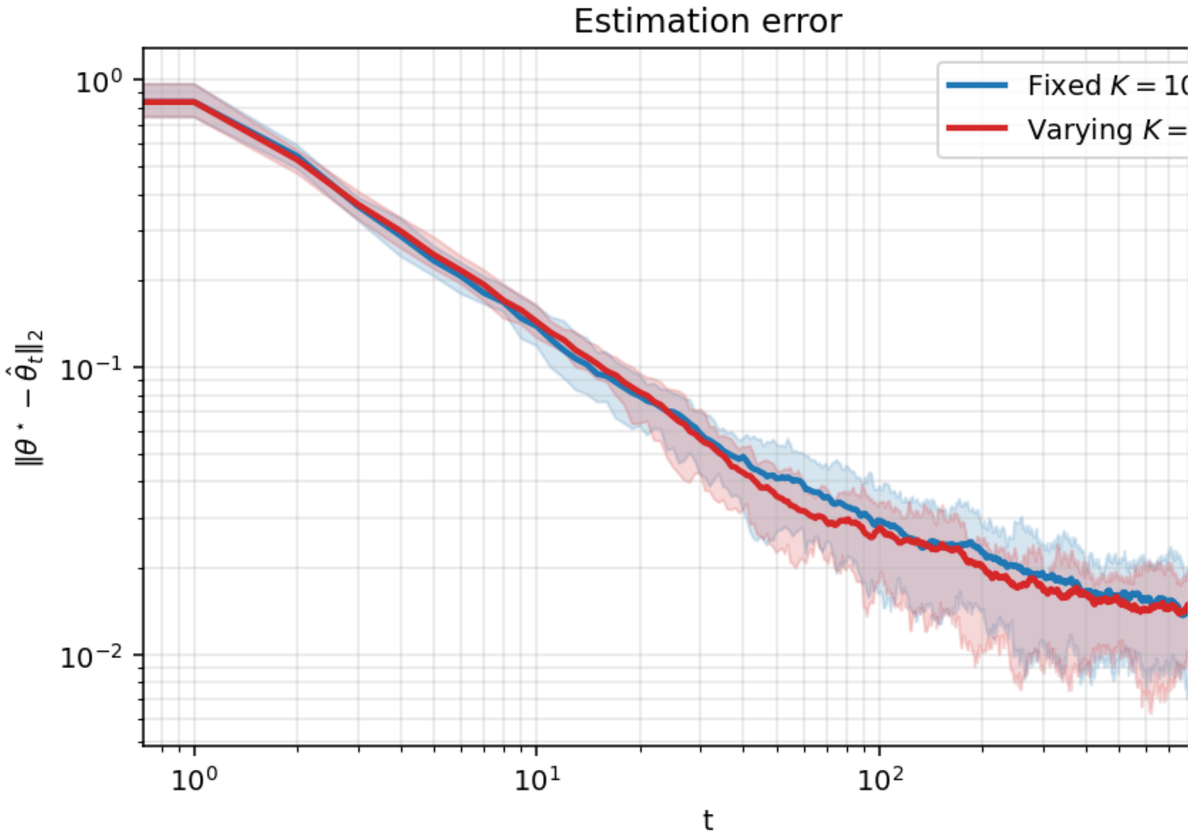

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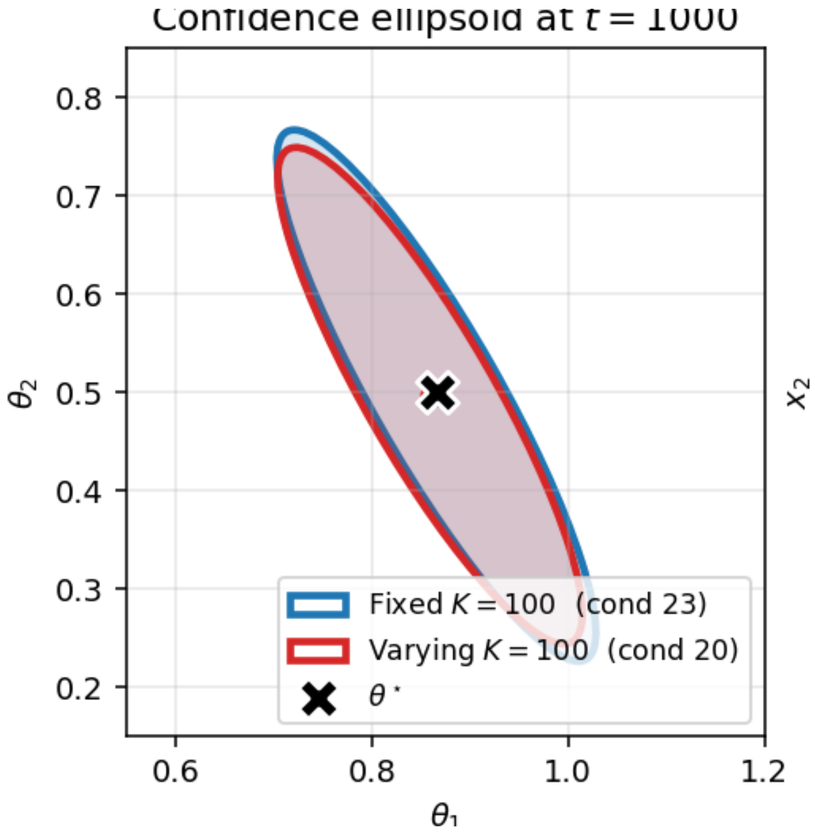

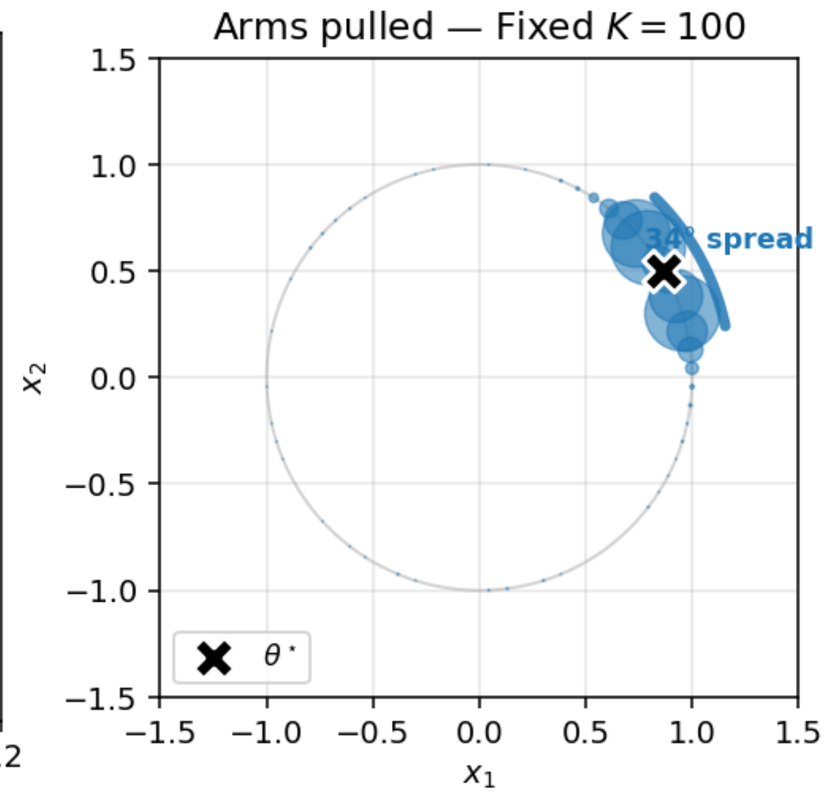

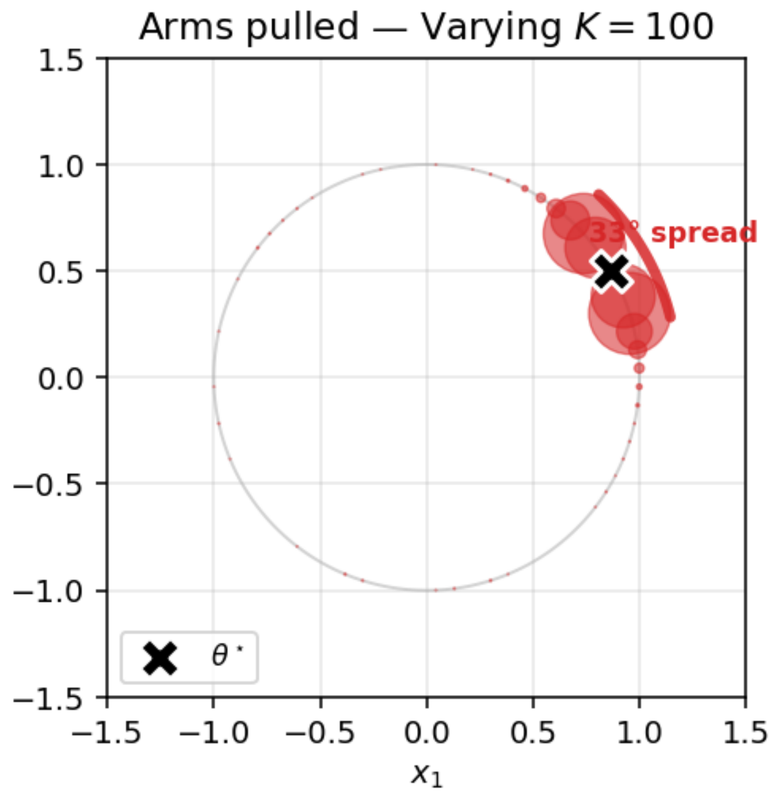

Can Linear Bandits Give Personalised Recommendations?

Benefits of Personalisation

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- In reality, user preferences vary greatly
  - Some may prefer sports, others may love politics
- How can we bring in user identity into the bandit problem?

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- Option 3: Incorporate user features in x
    - $x_{t,i} = \varphi(\text{user at time } t, \text{item } i)$
    - One shared θ, interpreted as weights given to each feature

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    - Found that using past data to cluster users was better than using joint user-item feature vectors
- Joint features: Used by the Yahoo! News article recommendation paper (Li et al, 2009)
    - Today, we use foundation models for extracting user profiles as well

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- Modern day data science makes smart use of neural networks!
- We will explore a few more practical aspects about bandits in upcoming lectures